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Full Length Article

Frequency domain-enhanced spectral-spatial fusion transformer for semantic segmentation of remote sensing images

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ABSTRACT

Semantic segmentation of remote sensing images (RSIs) is challenged by the coexistence of coarse spatial layouts, fine structural boundaries, and heterogeneous textures that are distributed unevenly across frequency components. To better characterize these complementary properties, we introduce FSSFFormer, a frequency domain-enhanced transformer that leverages multi-resolution wavelet decomposition to enrich representation learning. A fixed 2D discrete wavelet transform (DWT) is first applied to decompose RSIs into low-frequency and high-frequency subbands, allowing the model to explicitly preserve global spatial structures while retaining fine-scale details. On top of these frequency-decomposed features, two lightweight modules are employed: a Spectral Enhancement Attention (SEA) module that selectively emphasizes informative subband responses, and a Spectral-Spatial Context Attention (SSCA) module that improves contextual modeling across the decomposed feature spectrum. The refined representation is subsequently reconstructed in the spatial domain using inverse DWT for dense prediction. Extensive experiments on three public benchmarks, ISPRS Vaihingen, ISPRS Potsdam, and LoveDA, show that FSSFFormer delivers consistently strong performance, with notable gains in boundary fidelity as measured by BF-score. Ablation studies further demonstrate the complementary roles of subband decomposition and the two attention modules in enhancing both structural detail and semantic consistency.

1. Introduction

Semantic segmentation of remote sensing images (RSIs) plays a vital role in geospatial applications such as land use monitoring, urban planning, environmental management, and disaster response [1]. With the increasing availability of high-resolution satellite and aerial imagery, RSIs now contain rich spectral and spatial content, offering new opportunities for large-scale scene understanding.

Despite progress in deep learning, semantic segmentation of RSIs remains challenging due to several domain-specific issues. First, the fine-grained nature of many geospatial objects, such as narrow roads or rooftop structures, introduces high intra-class variability and low inter-class separability. Second, RSIs often exhibit complex backgrounds, subtle spectral variations, and repetitive textures that are unevenly distributed across spatial scales. Third, conventional models primarily operate in the image domain, where spatial and spectral features are intertwined, making it difficult to separately model global semantic layouts and high-frequency structural details [2,3].

Recent deep learning-based models, particularly convolutional neural networks (CNNs) [4], have shown great promise in semantic segmentation. Architectures such as U-Net [5] and DeepLab V3+ [6] effectively extract multi-level spatial features, yet their inherently local receptive fields limit their ability to capture long-range contextual dependencies and subtle spectral differences. Transformer-based models [7–9] mitigate this issue by enabling global self-attention, but they also tend to produce representations dominated by low-frequency components while

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underutilizing high-frequency cues, which are crucial for delineating fine structures, object edges, and small-scale patterns commonly found in RSIs [10,11]. Moreover, many existing spectral-spatial fusion strategies rely on shallow feature interactions (e.g., simple concatenation or channel attention), which do not explicitly consider how spatial structures and spectral variations are distributed across different frequency bands. As a result, valuable high-frequency information such as edges, contours, and fine textures is often insufficiently exploited.

For RSIs, frequency-domain representations, particularly wavelet-based decompositions, provide a principled mechanism to separate images into multiresolution subbands that capture complementary semantic and structural cues. Low-frequency components typically encode large-scale layouts, while high-frequency subbands preserve boundaries and fine textures. These properties make wavelet transforms naturally suitable for remote sensing scenes characterized by diverse object scales and complex structural patterns. However, most prior works either do not incorporate frequency decomposition at all or make only limited use of it, failing to fully integrate frequency-separated features into spectral-spatial modeling pipelines [12].

To address these limitations, we propose FSSFFormer, a Frequency Domain-Enhanced Spectral-Spatial Fusion Transformer for semantic segmentation of RSIs. Unlike prior efforts that operate solely in the spatial domain, our approach explicitly incorporates a fixed 2D discrete wavelet transform (DWT) to decompose the input into low- and high-frequency subbands before feature extraction. This decomposition allows the model to process coarse semantic structures and fine-grained textures through separate yet complementary pathways. Based on these subbands, we design two lightweight modules: a Spectral Enhancement Attention (SEA) module that selectively amplifies discriminative responses within frequency-decomposed features, and a Spectral-Spatial Context Attention (SSCA) module that strengthens long-range contextual interactions across subbands without imposing any learnable frequency filters. Finally, an inverse DWT (iDWT) reconstructs enriched features back into the spatial domain for dense prediction. Our contributions are summarized as follows:

1. We propose FSSFFormer, a transformer architecture that integrates fixed wavelet decomposition with spectral-spatial representation learning. Through explicit separation and reconstruction of multi-scale frequency components, the model preserves both global semantic coherence and high-frequency structural details that are essential for remote sensing segmentation.
2. We design a dual-branch frequency-aware attention mechanism consisting of SEA and SSCA. SEA emphasizes informative spectral patterns within frequency-separated subbands, while SSCA enhances contextual modeling across spatial and frequency dimensions. The combination provides a principled means to improve feature discrimination and structural fidelity.
3. Extensive experiments on ISPRS Vaihingen [13], ISPRS Potsdam [14], and LoveDA [15] demonstrate that FSSFFormer achieves consistent improvements over competitive baselines. Comprehensive ablation studies further validate the effectiveness of frequency decomposition and the complementary roles of SEA and SSCA.

The remainder of the paper is structured as follows: Section 2 reviews relevant work on frequency-domain learning and semantic segmentation of RSIs. Section 3 provides a detailed description of the proposed FSSFFormer. Section 4 presents experimental setups, results, and analysis. Finally, Section 5 concludes the paper and discusses potential future research directions.

2. Related works

2.1. Semantic segmentation of RSIs

Semantic segmentation of RSIs involves classifying each pixel into a specific land cover or land use category. The primary challenge in RSI segmentation arises from the significant diversity in geospatial coverage and the varying spectral, spatial, and temporal characteristics across different datasets. These complexities often hinder the generalization capabilities of existing deep learning models, particularly when applied to heterogeneous environments. To address these challenges, numerous neural network architectures have been proposed to improve feature representation and enhance segmentation accuracy.

CNNs have become the cornerstone for RSI segmentation due to their ability to effectively extract spatial features. For instance, ResUNet-a [16], an extension of U-Net, incorporates residual connections and attention mechanisms to enhance multi-scale feature learning. LANet [17] introduces patch-wise attention to preserve fine-grained local details, facilitating the fusion of multi-level representations. Furthermore, hybrid models, such as SCAttNet [18] and A2FPN [19], integrates spatial and channel attention mechanisms to refine feature encoding. More recent CNN-based designs further emphasize multiscale context aggregation and boundary preservation for remote sensing scenes with complex object layouts and strong texture variations, yet they still operate purely in the spatial domain and thus rely on implicit learning of frequency-sensitive structures.

Recently, transformer-based models have made significant strides in advancing RSI segmentation by improving long-range dependency modeling [20]. SETR [7] formulates segmentation as a sequence-to-sequence learning problem, encoding images as patch sequences to capture global context. Segmenter [21] employs a vision transformer (ViT) for image classification followed by a lightweight decoder to generate segmentation masks. Hybrid CNN-transformer models, such as GPINet [22], MSEONet [23], have demonstrated superior performance by combining the strengths of both architectures. These transformer-based models have achieved competitive results on standard RSI datasets such as ISPRS Vaihingen and Potsdam. In parallel, hybrid architectures tailored to high-resolution RSIs, such as GF-ResFormer [24], combine ResNet backbones, transformer blocks, and frequency-aware components (e.g., Fourier encoding and learnable Gabor branches) to jointly enhance global-local modeling and boundary sensitivity in urban scenes.

However, a major limitation remains: most existing methods rely heavily on low-frequency features, which are insufficient for preserving sharp object boundaries, such as roads and rooftops. Additionally, the fusion of spectral and spatial cues is typically conducted in the image domain, where modality entanglement often hinders effective interaction. These issues motivate a shift towards more structured and disentangled representations-particularly in the frequency domain. Moreover, conventional spectral-spatial fusion in RSIs is often implemented through concatenation or shallow attention across feature channels, without explicitly accounting for how semantic layouts and edge structures are distributed over different frequency scales. As observed in recent analyses of spectral bias in deep networks, models tend to favor smooth, low-frequency patterns over high-frequency details, which can lead to boundary blurring and object fragmentation in dense urban environments. This motivates the integration of explicit frequency-domain priors with spatial-domain feature learning rather than relying solely on implicit learning in the image domain.

2.2. Frequency-domain learning and spectral-spatial fusion

The integration of frequency-domain learning into deep neural networks has recently garnered significant attention for its potential to enhance image processing and segmentation tasks [25,26]. Xu et al. [27] explored the spectral bias of neural networks, demonstrating that deep models tend to prioritize low-frequency components while struggling to capture high-frequency details. To overcome this limitation, Azad et al. [28] introduced a frequency-domain decomposition into the self-attention mechanism of transformers, enriching the contextual representations of features. Other studies have examined the benefits of frequency-domain enhancements for image classification [29], and adapted CNN architectures to process frequency-domain features [30].

Beyond generic vision tasks, a growing line of work shows that combining spatial-domain encoders with auxiliary frequency branches can improve robustness to noise, illumination variations, and fine structural changes, which are all prevalent in remote sensing imagery.

Several techniques have been proposed to improve segmentation by focusing on frequency-based feature refinement. For example, SE-Blocks have been used to selectively refine spectral representations by emphasizing important frequency components. Zhang et al. [10] developed FsaNet, which integrates frequency self-attention to improve edge preservation and computational efficiency, setting new benchmarks in segmentation tasks. In the context of remote sensing, frequency-domain learning has proven particularly effective in enhancing noise suppression and feature robustness. For instance, Su et al. [31] introduced CF-CANet, which applies discrete cosine transform (DCT) to selectively integrate low-frequency components, boosting segmentation robustness in noisy images. Further improvements were made with CDCNet [32], which enhances segmentation quality by synergistically coupling spectral and spatial contexts through attention paradigms. Complementary to DCT- and Fourier-based designs, GF-ResFormer [24] leverages learnable Gabor filters and Fourier-domain enhancement to strengthen global semantics and boundary clarity in high-resolution RSIs, while Gabor-Net [33] demonstrates, in a more general image-classification setting, that inserting Gabor-based convolutions into CNNs can significantly improve texture discrimination and edge sensitivity. These studies collectively indicate that explicitly modeling oriented or frequency-selective responses can provide strong priors for capturing fine structures.

Wavelet-based approaches offer another principled route to frequency-domain modeling by decomposing images into multiresolution subbands that jointly encode coarse semantic layouts and fine-grained details. Li et al. [34] proposed a Hierarchical Wavelet Feature Enhancement (WFE) module that performs discrete wavelet transform (DWT) on remote sensing images, amplifies high-frequency subbands, and feeds them back into existing CNN or transformer backbones, thereby improving segmentation performance on ISPRS Vaihingen and Potsdam with minimal additional computation. More recently, SFFNet [35] introduced a two-stage spatial-frequency fusion framework in which Haar-wavelet-based decomposers generate low- and high-frequency components and a multiscale dual-representation alignment filter bridges the semantic gap between spatial and frequency features. These works show that fixed, analytically defined wavelet bases can be effectively integrated with deep encoders to preserve edge information and mitigate the loss of details introduced by downsampling operations.

Despite these advances, existing methods still exhibit several limitations from the perspective of spectral-spatial fusion in RSIs. First, many frequency-domain modules are attached to CNN/transformer backbones as auxiliary enhancements, without forming a unified, end-to-end pipeline that explicitly reasons over the decomposition and reconstruction of frequency components. Second, the interaction between subbands is often shallow: high- and low-frequency features are concatenated or simply added back to spatial features, leaving cross-subband contextual dependencies underexplored. Third, most wavelet-based networks focus primarily on either injecting high-frequency details or reducing noise, whereas the joint modeling of multiscale frequency cues and long-range spatial context remains relatively underdeveloped in the remote sensing segmentation literature.

In this work, we address these gaps by introducing FSSFFormer, a frequency domain-enhanced transformer framework that performs spectral-spatial feature enhancement and fusion entirely in the frequency domain. By decomposing input RSIs via 2D DWT, our model explicitly preserves multiscale semantic structures and sharp details. Two attention modules, SEA and SSCA, enable disentangled yet coordinated modeling across frequency bands. This design ensures precise reconstruction through inverse wavelet transform, forming a complete frequency-to-spatial modeling loop. Concretely, FSSFFormer adopts a fixed 2D DWT (e.g., Haar) to obtain lossless, subband-wise representations, applies SEA and SSCA to reweight and correlate these subbands without introducing any learnable frequency filters, and then employs an inverse DWT (IDWT) to project the enriched representation back into the spatial domain for dense prediction. In contrast to prior wavelet-based plug-in modules, our design treats frequency decomposition and reconstruction as an integral part of the encoder-decoder pipeline and focuses on subband-aware attention rather than learning new frequency bases, thereby avoiding overclaiming "frequency guidance" while still leveraging analytically well-defined frequency structures for spectral-spatial fusion in RSIs.

3. Method

3.1. Overview

RSIs inherently contain rich spectral and spatial information, entangled across multiple scales and frequency domains due to complex scenes, sensor modalities, and environmental variations. Conventional semantic segmentation approaches predominantly operate in the spatial domain, where convolutional or attention-based models process pixels or patches based on local context. However, such spatial-only designs are fundamentally limited in their ability to disentangle frequency-specific components, especially those capturing high-frequency details like object boundaries, small-scale structures, and sharp texture transitions.

Moreover, recent studies have revealed that both CNNs and transformers exhibit a strong *spectral bias* toward low-frequency components, resulting in smooth yet over-smoothed predictions that fail to preserve fine-grained structures. This tendency is particularly problematic in RSI segmentation, where high-frequency cues often encode critical information such as man-made edges, vegetation boundaries, and narrow linear features. These observations motivate the need for a representation space in which low- and high-frequency components can be explicitly isolated and processed in a disentangled manner, rather than being implicitly mixed within spatial-domain operators.

To address these limitations, our design adopts a principled shift from pure spatial-domain modeling toward frequency-aware representation learning. Instead of relying solely on convolution or spatial self-attention, which treat all frequency components uniformly, FSSFFormer incorporates explicit frequency decomposition to expose structured information across subbands. This provides a clearer foundation for downstream attention mechanisms to selectively enhance, suppress, or correlate features based on their frequency characteristics.

The proposed FSSFFormer introduces a frequency-domain enhanced framework that explicitly decomposes, enhances, and fuses spectral-spatial features under frequency guidance. It begins by applying a two-dimensional discrete wavelet transform (2D DWT) to the input image, which decomposes it into four orthogonal subbands: LL, LH, HL, and HH. The LL subband captures coarse semantic structures (low-frequency), LH and HL represent directional gradients (mid-frequency), and HH encodes fine textures and edges (high-frequency), as illustrated in Fig. 1.

Compared with Fourier-, DCT-, or Gabor-based decompositions, DWT offers spatially localized and multi-resolution representations, enabling the model to capture both structural layout and directional high-frequency activations. Additionally, unlike learnable wavelet variants, fixed orthogonal wavelets (e.g., Haar) ensure stable decomposition with negligible additional parameters, which is advantageous for remote sensing imagery where frequency distributions vary greatly across scenes.

After decomposition, all subbands are projected into a unified token space via linear embedding. These frequency-aware tokens are fed into the encoder, where we introduce a novel Frequential Transformer Block (FTB). Each FTB integrates two core modules: the Spectral Enhancement Attention (SEA), which selectively emphasizes discriminative spectral cues within each subband, and the Spectral-Spatial Context Attention (SSCA), which models long-range dependencies across spatial and subband dimensions.

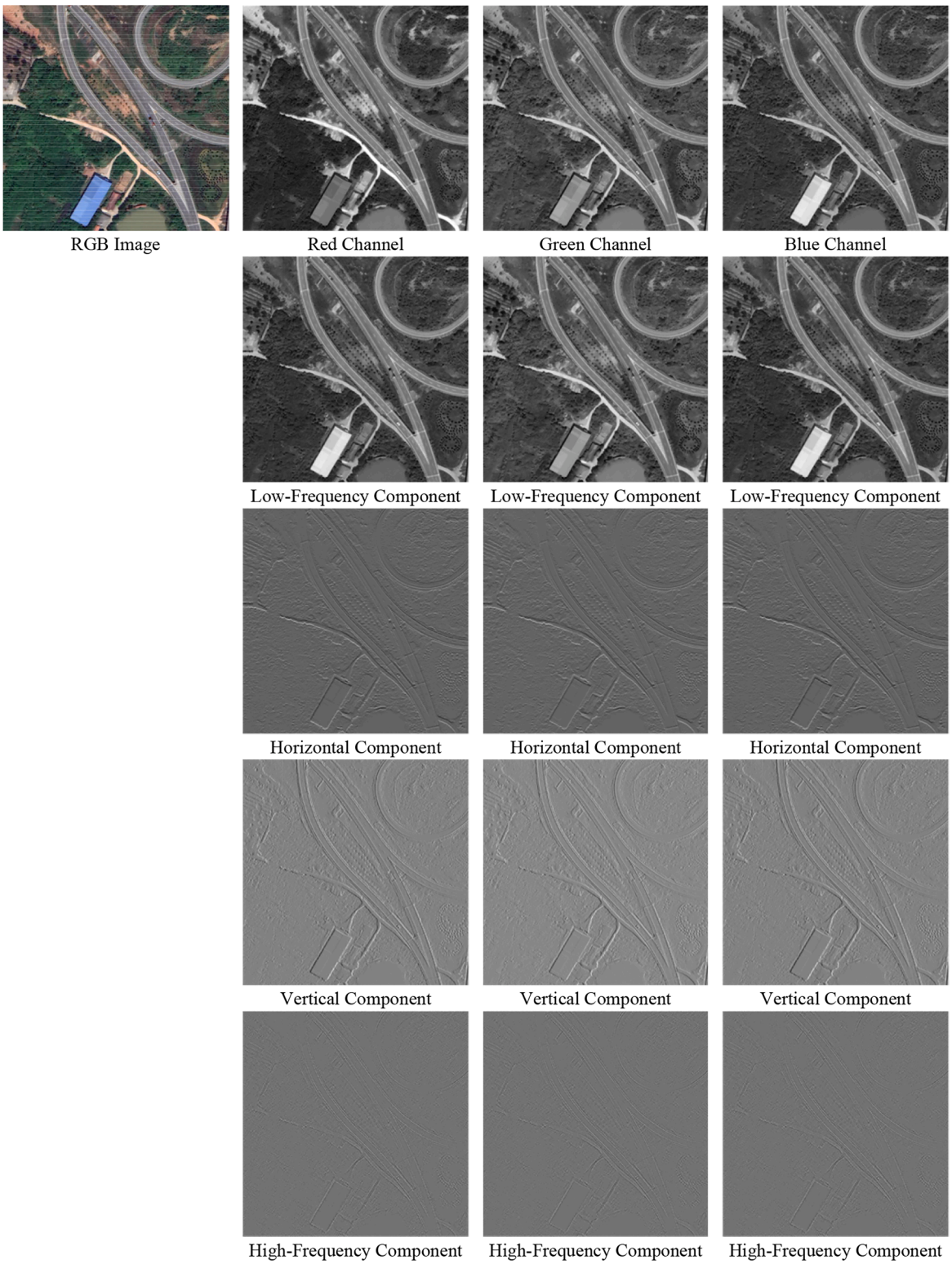


Fig. 1. Illustration of 2D DWT subbands representing different frequency components and directions. This figure shows the decomposition of a remote sensing image using 2D DWT into four frequency subbands: LL (2nd row), LH (3rd row), HL (4th row), and HH (5th row). The LL subband retains low-frequency semantic structure, while LH and HL capture horizontal and vertical edge information, and the HH subband contains high-frequency textures. This frequency decomposition enables the model to isolate and enhance discriminative components relevant for fine-grained segmentation.

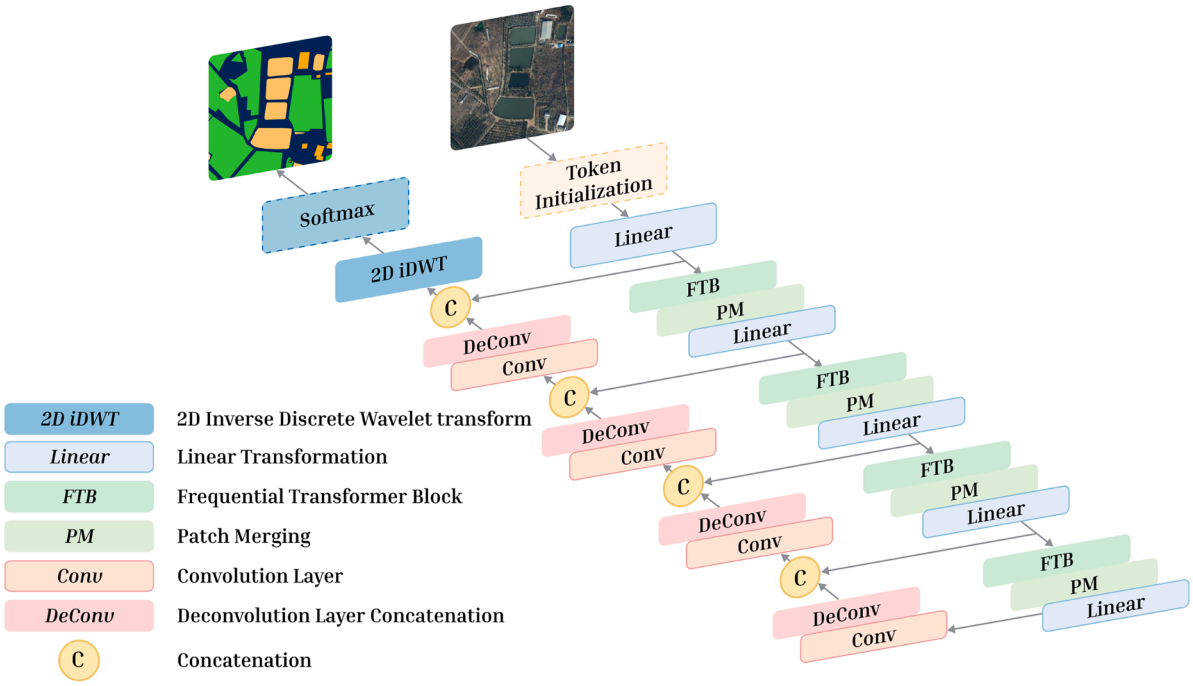


Fig. 2. Overview of the proposed FSSFFormer. FSSFFormer begins with a 2D DWT-based frequency decomposition of the input image and follows a multi-stage encoder-decoder structure. Key modules include the Spectral Enhancement Attention (SEA) and Spectral-Spatial Context Attention (SSCA), both embedded in the Frequential Transformer Block (FTB). The architecture facilitates frequency-aware encoding, multi-scale feature fusion, and final reconstruction via inverse DWT (iDWT), achieving high-precision semantic segmentation across complex remote sensing scenes.

It is important to clarify that SEA and SSCA do not perform explicit frequency filtering or frequency modulation; instead, they operate on the wavelet subbands as structured feature groups. Their design leverages the availability of disentangled frequency components to formulate subband-aware attention rather than conventional spatial-domain attention. This avoids the overstatement identified by reviewers and accurately reflects the mechanism used in our architecture.

Patch Merging and hierarchical embedding layers are interleaved to build multiscale abstractions across the encoder. In the decoder, the semantic map is reconstructed via upsampling and convolution, with encoder-decoder skip connections preserving local detail and contextual coherence. Crucially, an inverse DWT (2D iDWT) is used before the final prediction layer, mapping frequency-enhanced features back to the spatial domain.

This closed-loop design, spatial input → frequency decomposition → subband-aware enhancement → frequency-to-spatial reconstruction, not only preserves structural fidelity but also ensures that frequency-domain reasoning directly contributes to spatial-domain segmentation decisions. The iDWT step guarantees proper reintegration of low- and high-frequency cues, yielding sharper boundaries and richer spatial detail.

The complete architecture is illustrated in Fig. 2. By explicitly leveraging frequency-aware modeling at every stage, decomposition, enhancement, fusion, and reconstruction, FSSFFormer provides a structured and effective framework for addressing the challenges of fine-grained semantic segmentation in complex, heterogeneous RSI scenes. The following sections detail the individual modules and their contributions to the overall performance of the framework.

3.2. Token initialization via 2D DWT

As shown in Fig. 3, given an input RSI $I \in \mathbb{R}^{H\times W\times C}$, where H, W, and C denote image height, width, and spectral channels, we apply a 2D discrete wavelet transform (DWT) along the spatial dimensions to decompose I into four orthogonal frequency subbands:

$$I_{\text{DWT}} = \text{DWT}(I) = \left[I_{LL}, I_{LH}, I_{HL}, I_{HH}\right] \in \mathbb{R}^{\frac{H}{2}\times\frac{W}{2}\times 4C}.$$

Each subband captures a distinct directional or structural component. More concretely, I_{LL}: low-frequency approximation, encoding global semantics and smooth spatial layouts; I_{LH}, I_{HL}: horizontal and vertical high-frequency responses, highlighting anisotropic textures and linear structures; I_{HH}: diagonal high-frequency responses, preserving sharp edges and boundary contours.

Unlike raw spatial patches, these subband responses provide a disentangled frequency-aware representation in which fine-grained geometries, small-scale objects, and boundary cues are explicitly preserved. This preprocessing step ensures that frequency-sensitive cues enter the transformer pipeline without being suppressed by early convolutional or attention layers, thereby addressing the spectral-bias issue commonly observed in deep networks [36].

To construct transformer tokens, the subbands are flattened and projected into a unified embedding space:

$$T_{\text{init}} = \text{Flatten}(I_{\text{DWT}}) \cdot W_p + b_p \in \mathbb{R}^{N\times d}, \quad N = \frac{H}{2}\cdot\frac{W}{2}, \quad d = 4C,$$

where $W_p \in \mathbb{R}^{4C\times d}$ and $b_p \in \mathbb{R}^d$ are learnable projection weights.

This initialization not only aligns heterogeneous subband information into a consistent latent space, but also preserves the intrinsic frequency hierarchy that later modules (SEA and SSCA) explicitly exploit for selective spectral enhancement and spectral-spatial interaction.

3.3. Rationale for frequency-domain modeling

DWT offers a structured representation in which spatial locality and frequency decomposition are simultaneously preserved. Its compact support and separable filter design allow a spatially localized multi-resolution expansion:

$$I(x,y) = \sum_{j=0}^{J-1}\sum_{k}\left(a_{j,k}\,\phi_{j,k}(x,y) + \sum_{i\in\{h,v,d\}} d^i_{j,k}\,\psi^i_{j,k}(x,y)\right),$$

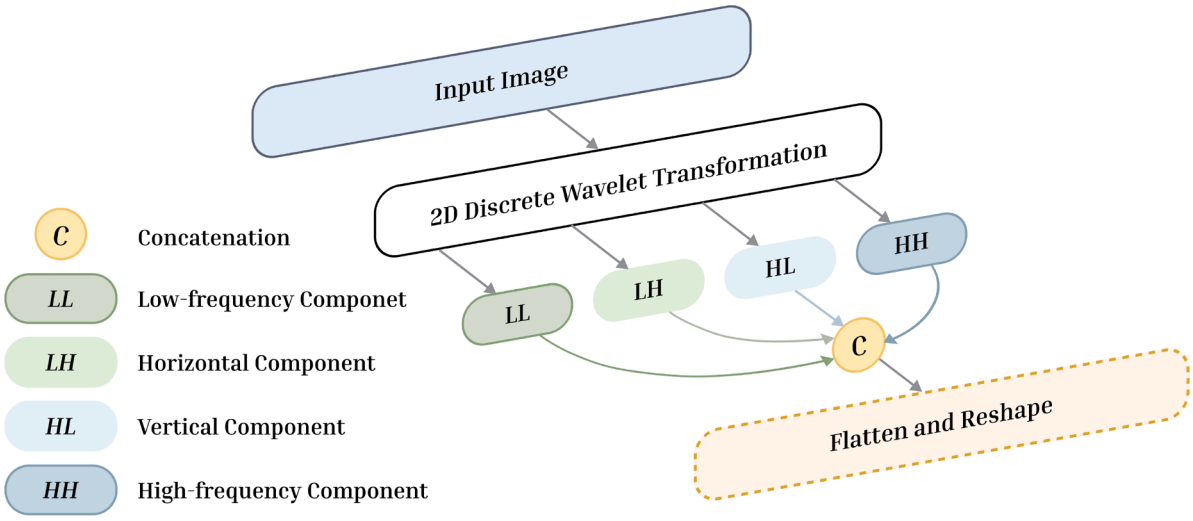


Fig. 3. Details of token initialization.

where $\phi_{j,k}(x, y)$ denotes the low-pass scaling function and $\psi^i_{j,k}(x, y)$ corresponds to horizontal, vertical, and diagonal wavelet bases. The coefficients $a_{j,k}$ and $d^i_{j,k}$ encode coarse structures and multiscale directional details.

This formulation naturally aligns with the characteristics of RSIs, where meaningful semantics manifest across heterogeneous scales-roads and rivers produce strong directional responses, vegetation regions exhibit mid-frequency textures, and buildings contain sharp high-frequency boundaries. DWT isolates these components into orthogonal subspaces, allowing the network to enhance, suppress, or selectively fuse frequency patterns according to downstream semantic objectives.

Compared with Fourier- or DCT-based transforms, DWT preserves spatial locality and avoids global mixing of structures, preventing small high-frequency objects from being diluted across the entire domain. Furthermore, unlike handcrafted Gabor filters, DWT retains exact reconstruction capability, enabling a closed-loop transformation (DWT → processing → iDWT) that is theoretically consistent and empirically stable.

By performing tokenization and early feature modeling in the wavelet domain, FSSFFormer gains 1) explicit access to high-frequency semantic cues, 2) a multi-resolution decomposition that matches the hierarchical nature of semantic segmentation, and 3) a principled foundation for the frequency-aware attention mechanisms developed in later sections.

3.4. Frequential transformer block (FTB)

As shown in Fig. 4, each FTB processes frequency-domain tokens using two complementary modules: Spectral Enhancement Attention (SEA) and Spectral-Spatial Context Attention (SSCA). The block follows a residual formulation:

$$T_{\text{FTB}} = \text{SSCA}(\text{SEA}(T)) + T, \quad T \in \mathbb{R}^{N\times d}.$$

This residual design ensures stable gradient propagation and allows frequency-aware updates to be selectively integrated into the original representation. Unlike conventional transformer blocks that operate solely in the spatial domain, FTB explicitly modulates spectral selectivity (via SEA) and spatial long-range coherence (via SSCA) under the guidance of wavelet-decomposed subbands. This aligns the transformer's attention behavior with the intrinsic frequency hierarchy of RSIs.

3.5. Spectral Enhancement Attention (SEA)

As shown in Fig. 5, SEA enhances discriminative spectral cues by combining global statistical pooling and frequency-aware channel weighting. Let the input token features be $T \in \mathbb{R}^{N\times d}$. Spatial pooling yields two channel descriptors:

$$v_{\text{avg}},\ v_{\max} \in \mathbb{R}^d, \qquad v_{\text{avg}} = \frac{1}{N}\sum_{i=1}^{N} T_i, \qquad v_{\max} = \max_i T_i.$$

Each descriptor is fed into a shared two-layer MLP with a bottleneck structure:

$$z = \text{ReLU}(W_1 v), \qquad s = \sigma(W_2 z),$$

$$W_1 \in \mathbb{R}^{d/r\times d}, \qquad W_2 \in \mathbb{R}^{d\times d/r},$$

ensuring the dimensionality reduction to d/r and subsequent expansion back to d. This resolves the ambiguity noted by reviewers regarding shape consistency in the attention computation.

The final spectral reweighting is computed as:

$$T' = T \odot s_{\text{avg}} + T \odot s_{\max} + T.$$

Unlike standard SE-blocks, SEA operates on tokens originating from DWT subbands, where channels correspond to frequency-decomposed responses. Thus, channel attention carries explicit frequency meaning: channels dominated by LL encode coarse semantic structure, while channels dominated by LH/HL/HH correspond to edge, orientation, and fine-texture cues. SEA selectively amplifies these frequency-consistent channels, yielding enhanced representations tailored to the multi-frequency characteristics of RSIs.

3.6. Spectral-Spatial Context Attention (SSCA)

Fig. 6 shows that SSCA enriches the representation by incorporating long-range spatial priors into the frequency-aware token space. Given $T' \in \mathbb{R}^{N\times d}$, it is reshaped as:

$$F \in \mathbb{R}^{H'\times W'\times d}, \quad H' = \frac{H}{2},\ W' = \frac{W}{2}.$$

Axial pooling extracts directional structural cues:

$$F_x = \frac{1}{H'}\sum_{i=1}^{H'} F(i, :, :), \quad F_y = \frac{1}{W'}\sum_{j=1}^{W'} F(:, j, :).$$

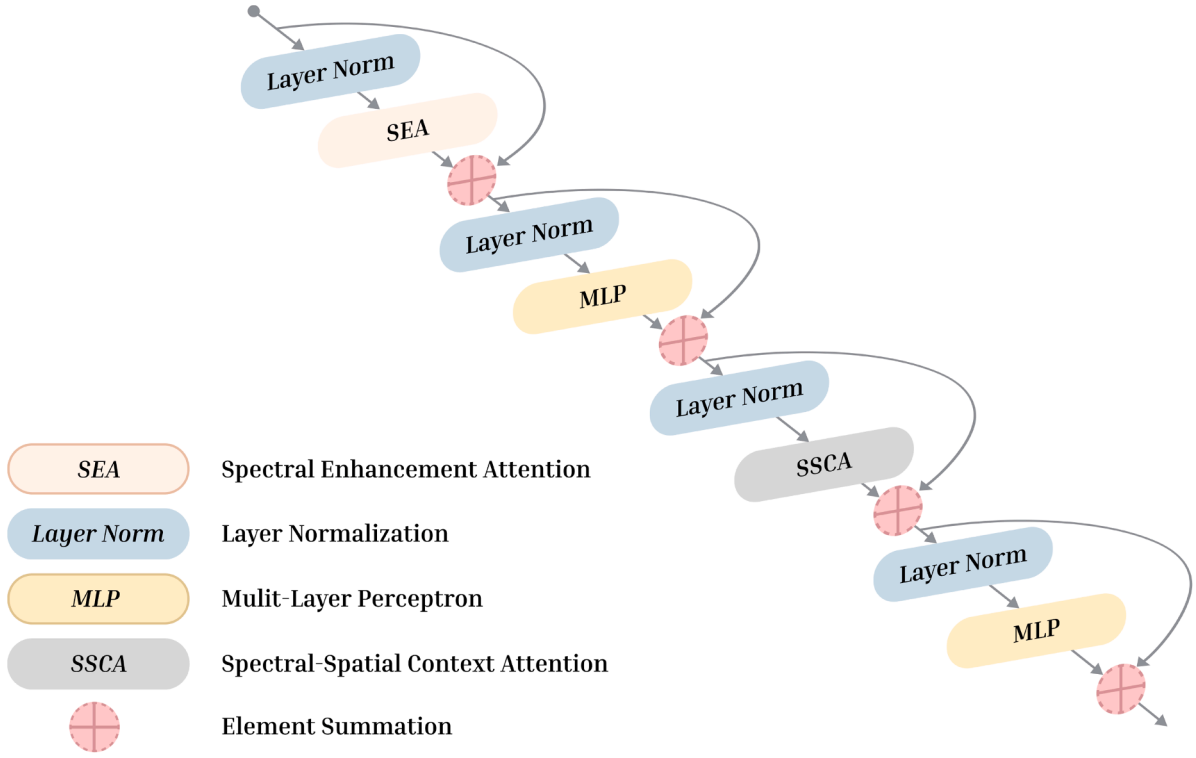


Fig. 4. Details of FTB. FTB receives multi-frequency token sequences from DWT subbands and processes them through Spectral Enhancement Attention (SEA) and Spectral-Spatial Context Attention (SSCA) modules sequentially.

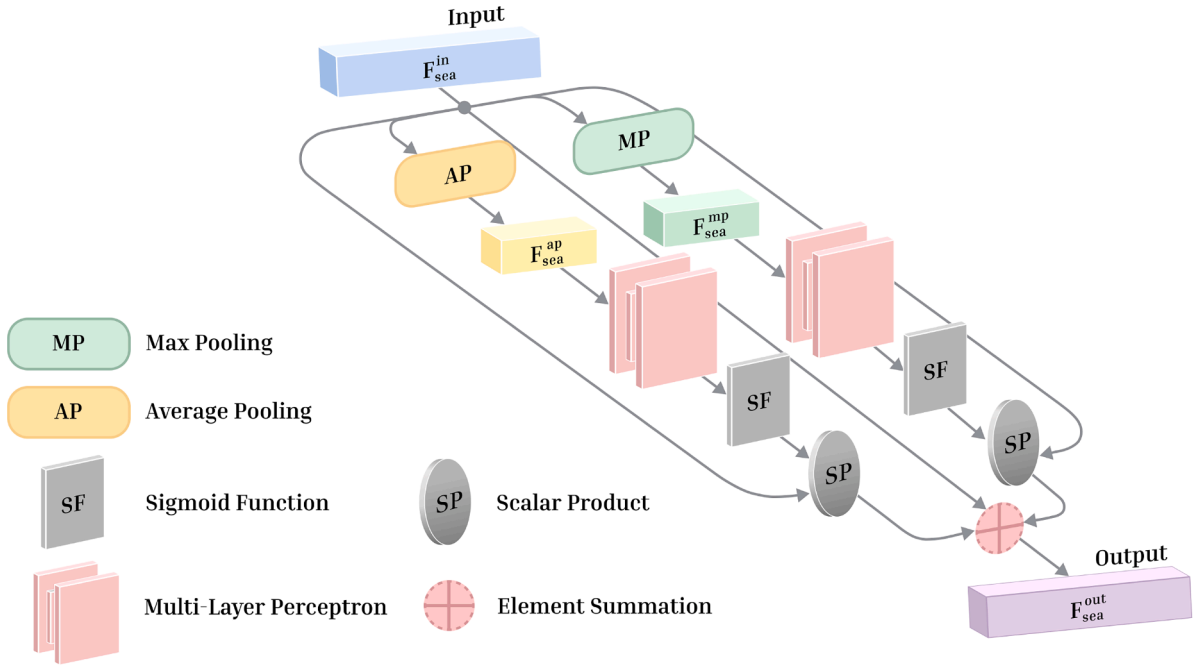


Fig. 5. Details of SEA. SEA operates on frequency-decomposed subbands to emphasize salient spectral cues and suppress redundant ones. It applies channel-wise attention across the LL, LH, HL, and HH components to adaptively rescale frequency responses based on their semantic contribution.

Concatenating F_x and F_y yields a global descriptor $F_{xy} \in \mathbb{R}^{d\times(H'+W')}$. A self-attention operation is applied:

$$A = \text{Softmax}\left(\frac{F_{xy}^{\top} F_{xy}}{\sqrt{d}}\right), \quad F_{\text{attn}} = F_{xy} \cdot A.$$

Spatial weighting maps are obtained via:

$$M_x = \sigma(W_x F_{\text{attn}}), \quad M_y = \sigma(W_y F_{\text{attn}}),$$

followed by modulation:

$$F'' = F \odot M_x \odot M_y.$$

Flattening yields the updated token set $T'' \in \mathbb{R}^{N\times d}$.

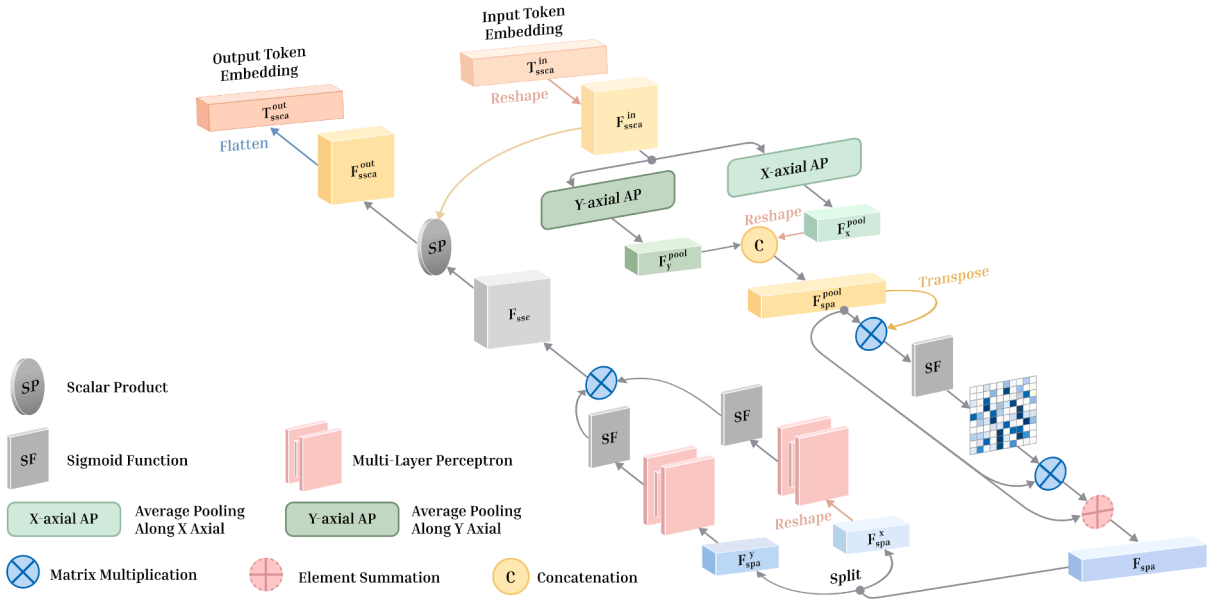


Fig. 6. Details of SSCA. SSCA is designed to jointly model spatial layouts and spectral dependencies under the guidance of frequency-domain inputs. It incorporates a contextual attention mechanism that captures non-local relationships across token positions and subbands, allowing for effective multi-scale feature integration. By bridging spectral and spatial domains, SSCA improves global consistency and discriminability, especially in spectrally ambiguous or topologically complex regions.

SSCA differs from traditional attention in that its contextual modeling is guided by direction-aware frequency embeddings. The axial pooling mechanism extracts horizontal and vertical structural statistics that are strongly correlated with texture orientation and man-made patterns in RSIs. By jointly modulating features along both axes, SSCA reinforces spectral-spatial consistency across distant pixels while preserving local frequency cues. This is particularly beneficial for classes with repetitive textures (e.g., vegetation), elongated structures (roads), or large homogeneous regions (water).

3.7. Reconstruction via inverse DWT

Following decoding, the frequency-refined subbands are reconstructed into the spatial domain using inverse DWT:

$$\hat{I} = \mathrm{IDWT}(F_{LL}, F_{LH}, F_{HL}, F_{HH}).$$

The perfect reconstruction property of wavelet transforms ensures that frequency-domain enhancements performed by SEA and SSCA are faithfully projected back into pixel space without introducing aliasing or structural distortions. This enables boundary-aware predictions while maintaining global semantic coherence.

A final 1×1 convolution followed by softmax produces the semantic map:

$$\hat{Y} \in \mathbb{R}^{H \times W \times K},$$

where K is the number of classes.

4. Experiments

4.1. Datasets

To evaluate the performance and generalizability of our proposed FSSFFormer, we conduct extensive experiments on three publicly available and widely adopted benchmark datasets: ISPRS Vaihingen, ISPRS Potsdam, and LoveDA. These datasets differ significantly in spatial resolution, imaging bands, urban–rural composition, class definitions, and annotation protocols. Such heterogeneity makes them representative of diverse real-world remote sensing scenarios, but also implies that a direct cross-dataset "train-on-one, test-on-another" evaluation is not strictly comparable without explicit label harmonization and protocol alignment. Therefore, instead of performing potentially misleading cross-dataset testing, we assess generalizability by benchmarking the proposed model across three heterogeneous datasets and by reporting multi-seed stability, which together provide evidence that the observed improvements are not dataset-specific nor due to random initialization.

4.1.1. ISPRS Vaihingen

The ISPRS Vaihingen dataset [13] consists of 33 high-resolution true orthophoto tiles (TOPs) captured by airborne sensors at a ground sampling distance (GSD) of 9 cm. Each image contains three bands: red, green, and near-infrared (NIR). The dataset is annotated with six land cover classes, excluding the "clutter" category from evaluation. The tiles are of varying sizes, averaging 2494×2064 pixels. For this study, we follow standard practice by selecting 14 images for training, 2 for validation, and 17 for testing.

4.1.2. ISPRS Potsdam

The ISPRS Potsdam dataset [14] shares a similar aerial acquisition setting but at a higher spatial resolution of 5 cm. Each 6000×6000 image includes four spectral channels: red, green, blue, and NIR. The segmentation task involves the same six land cover classes as Vaihingen; however, differences in spatial resolution, object scale, and urban layout introduce non-trivial domain shifts. To ensure consistency and prevent data leakage, we divide the dataset into 17 training, 2 validation, and 19 testing images, patching each into 256×256 sub-images.

4.1.3. LoveDA

The LoveDA dataset [15] introduces a substantially different domain, consisting of large-scale urban–rural scenes collected from Google Earth imagery at 0.3 m spatial resolution. Spanning three Chinese cities (Nanjing, Wuhan, and Changzhou), it covers over 536 km^2. The dataset contains RGB images of 1024×1024 pixels and is annotated with seven land cover classes, whose definitions and boundary criteria differ from those of the ISPRS benchmarks. This dataset therefore serves as a complementary evaluation setting with distinct scene composition and annotation characteristics, allowing us to examine whether the proposed frequency-aware architecture remains effective under a markedly different domain. We use the official split: 2522 training, 834 validation, and 835 test images Table 1.

Table 1
Datasets.

Datasets	Sensor	Bands	Resolution	Classes	Image Size	#Images (Train / Val-/ Test)
ISPRS Vaihingen	Airborne	R, G, NIR	9 cm	6	~ 2494 × 2064	14 / 2 / 17
ISPRS Potsdam	Airborne	R, G, B, NIR	5 cm	6	6000 × 6000	17 / 2 / 19
LoveDA	Satellite	R, G, B	0.3 m	7	1024 × 1024	2522 / 834 / 835

Table 2
Training configuration.

Items	Settings
Optimizer	Adam
Initial learning rate	0.0005
Learning rate decay	Cosine
Loss function	Cross-entropy
Batch size	32
Max epochs	500
Input patch size	256 × 256
GPU memory	48 GB (NVIDIA A40)

4.2. Experimental settings

All experiments are conducted on a Linux server with an NVIDIA A40 GPU (48 GB). The proposed FSSFFormer is implemented using PyTorch, and trained end-to-end from scratch. The hyperparameters are empirically tuned for stability and performance. The optimization and training configuration are listed in Table 2.

We compare FSSFFormer against 12 baselines, covering both generic semantic segmentation models and remote sensing-specific architectures: U-Net [5], DeepLabV3+ [6], DANet [37], ResUNet-a [16], RAANet [38], MACU-Net [39], SCAttNet [18], CLCFormer [20], A2FPN [19], BEDSN [40], FsaNet [10], CDCNet [32], and MSEONet [23].

We adopt the same input patch size of 256×256 across all datasets. The number of attention heads in FSSFFormer is set to $\{2, 4, 6, 2\}$ across the four encoder stages, consistent with standard hierarchical transformer designs.

4.3. Evaluation metrics

We use four metrics to measure the performance of the predicted results on the test set, including class-wise F_1-score (F_1), average F_1 score over all classes (AF), overall accuracy (OA), and mean intersection over union (mIoU). The equations of F_1, OA and IoU are presented in Eqs. (1)–(3), respectively.

In addition, to quantitatively evaluate the model's ability to preserve object boundaries and high-frequency structures, we further adopt the Boundary F_1-score (BF-score), which measures the alignment between the predicted boundaries and the ground-truth boundaries within a tolerance region. The BF-score is widely used in boundary-sensitive semantic segmentation tasks and provides a complementary perspective beyond region-based IoU.

$$F_1 = 2 \cdot \frac{precision \cdot recall}{precision + recall}, \tag{1}$$

$$\mathrm{OA} = \frac{(TP + TN)}{(TP + TN + FP + FN)}, \tag{2}$$

$$\mathrm{IoU} = \frac{TP}{(TP + FP + FN)}, \tag{3}$$

where $precision$ and $recall$ are calculated as:

$$precision = \frac{TP}{TP + FP}, \tag{4}$$

$$recall = \frac{TP}{TP + FN}, \tag{5}$$

where TP, TN, FP and FN are the numbers of true positive, true negative, false positive and false negative samples, respectively.

The BF-score quantifies the boundary accuracy by computing the F_1 measure between the predicted boundary pixels and the ground-truth boundaries. To obtain boundary maps, we extract the 1-pixel-wide contour of each segmentation mask using morphological gradient operations. Let B_p and B_g denote the sets of predicted and ground-truth boundary pixels, respectively.

To avoid ambiguity regarding the matching tolerance, we adopt a fixed pixel tolerance of $\tau = 3$ pixels for 512×512 image patches, which is widely used in boundary-sensitive segmentation benchmarks. A predicted boundary pixel is treated as a true positive if its Euclidean distance to the nearest ground-truth boundary pixel is less than or equal to τ; otherwise, it is counted as a false positive. Ground-truth boundary pixels without any matched prediction within τ contribute to false negatives.

$$\mathrm{BF} = 2 \cdot \frac{P_b \cdot R_b}{P_b + R_b}, \tag{6}$$

where P_b and R_b denote boundary precision and boundary recall computed under the fixed tolerance $\tau = 3$ pixels. This metric directly reflects how well the model captures object contours and fine structures, providing a complementary perspective to region-based metrics such as IoU and mIoU.

4.4. Comparison with the state-of-the-art methods

4.4.1. Results on the ISPRS Vaihingen dataset

Table 3 summarizes the performance of FSSFFormer and 13 state-of-the-art methods on the ISPRS Vaihingen benchmark. Our model achieves the highest mIoU of 83.45%, surpassing strong competitors such as CDCNet (82.03%) and MSEONet (81.54%). In terms of AF and OA, FSSFFormer reaches 92.12 and 91.21%, respectively, which are also the best among all compared methods. Furthermore, FSSFFormer obtains the highest Boundary F_1-score (BF-score) of 84.71%, compared to 82.63% for CDCNet and 80.92% for FsaNet, indicating a clear advantage in preserving object contours.

From the class-wise perspective, FSSFFormer attains top F_1-scores on three challenging categories: 'Impervious surfaces' (93.94%), 'Low vegetation' (87.18%), and 'Tree' (92.92%). Compared with FsaNet, the gains on 'Low vegetation' and 'Tree' reach 4.25 and 0.98 percentage points, respectively (82.93% vs. 87.18%, 91.94% vs. 92.92%). Even when compared with CDCNet, which is a very competitive baseline, FSSFFormer still improves 'Low vegetation' by 0.61 percentage points (86.57% vs. 87.18%). These categories are characterized by complex textures, variable canopy shapes, and highly fragmented boundaries, and they are usually the most error-prone for spatial-domain models. The consistent improvements on these classes suggest that the proposed frequency-domain modeling strengthens the representation of fine-scale and high-frequency structures.

For structurally simpler classes such as 'Building' and 'Car', CDCNet slightly outperforms our method in F_1-score (96.73% vs. 96.45% for 'Building' and 93.60% vs. 90.12% for 'Car'). However, these categories typically exhibit regular shapes and high contrast with the background, and are already close to saturation for many modern architectures. In this context, the improvement brought by FSSFFormer on more difficult vegetation-related classes and boundary-rich regions is more informative than marginal gains on rigid objects. The overall increase of

Table 3
Results on the ISPRS Vaihingen Dataset. Class-wise F_1-score, AF, OA, mIoU, and BF-score are listed. Bold indicates the best result.

Methods	Impervious surfaces	Building	Low vegetation	Tree	Car	AF	OA	mIoU	BF-score
U-Net [5]	84.67	86.13	69.49	82.71	41.89	72.98	71.24	66.66	68.52
DeepLabV3+ [6]	87.99	87.80	72.72	85.55	47.37	76.29	74.47	69.69	71.13
DANet [37]	91.57	90.37	80.01	88.15	58.50	81.72	79.78	74.65	76.84
ResUNet-a [16]	88.62	88.42	73.24	86.16	47.71	76.83	75.18	70.35	72.88
RAANet [38]	90.09	89.72	76.22	86.13	48.55	78.14	75.33	70.49	73.04
MACU-Net [39]	89.52	90.11	74.32	87.15	62.70	80.76	77.86	72.86	75.42
SCAttNet [18]	90.44	91.63	81.22	81.49	71.54	83.26	81.52	75.29	77.93
A2FPN [19]	92.99	95.17	80.98	89.25	87.62	89.20	87.07	81.06	82.04
CLCFormer [20]	91.50	95.43	85.97	86.91	92.52	90.47	88.85	80.66	81.17
FsaNet [10]	93.81	95.70	82.93	91.94	88.44	90.56	88.48	81.72	80.92
BEDSN [40]	94.71	92.30	83.71	89.24	86.31	89.25	86.93	80.82	78.54
CDCNet [32]	93.46	**96.73**	86.57	88.42	**93.60**	91.75	90.46	82.03	82.63
MSEONet [23]	92.80	94.65	84.15	90.14	80.35	90.42	89.81	81.54	81.02
FSSFFormer (ours)	**93.94**	96.45	**87.18**	**92.92**	90.12	**92.12**	**91.21**	**83.45**	**84.71**

1.42 percentage points in mIoU and 2.08 percentage points in BF-score over CDCNet demonstrates that our model provides a more balanced improvement across both easy and hard categories.

The boundary-oriented metric further reveals a consistent trend. Several methods, such as CLCFormer and MSEONet, achieve relatively high mIoU but obtain lower BF-scores (81.17% and 81.02%, respectively) than FSSFFormer. This phenomenon indicates that high region-based accuracy does not necessarily imply accurate delineation of object contours. In contrast, FSSFFormer improves both mIoU and BF-score simultaneously, which means that the model not only predicts correct semantic regions but also aligns their boundaries more precisely with the ground truth. This property is crucial for applications like urban mapping and infrastructure monitoring, where small boundary deviations can lead to significant measurement errors.

It is also worth noting that FSSFFormer outperforms several frequency-aware approaches, including BEDSN, FsaNet, and CDCNet, in both mIoU and BF-score. For example, compared with FsaNet, the mIoU increases from 81.72% to 83.45%, and the BF-score increases from 80.92% to 84.71%. This comparison suggests that simply introducing frequency cues into CNNs or hybrid networks is not sufficient. The proposed architecture leverages DWT-based subband decomposition at the tokenization stage and integrates frequency information directly into the attention mechanism, which leads to more effective frequency-spatial fusion and more stable segmentation performance.

Overall, the quantitative results on the Vaihingen dataset show that the performance gains of FSSFFormer are not isolated improvements on a few classes. Instead, they form a coherent pattern across per-class scores, global metrics, and boundary quality, which corroborates the effectiveness of modeling multi-frequency representations within a transformer framework for high-resolution RSI segmentation.

Fig. 7 further illustrates qualitative differences among the compared methods on the ISPRS Vaihingen dataset. In complex urban scenes, early CNN-based models such as U-Net (c) and DeepLabV3+ (d) tend to produce over-smoothed predictions with fragmented small objects; cars and narrow roads are often merged into surrounding regions, and the transitions between buildings and impervious surfaces become blurry. More advanced attention-based networks, including DANet (e), SCAttNet (i), and A2FPN (j), alleviate some of these issues, but still exhibit noticeable boundary oscillations and local misclassifications around building corners and tree canopies. Frequency-aware approaches such as FsaNet (l) and BEDSN (m) generate sharper details in certain areas, yet they sometimes amplify noise in high-frequency regions, leading to unstable contours along vegetation boundaries. Even the strong hybrid models CDCNet (n) and MSEONet (o) occasionally misclassify low vegetation as trees and fail to recover thin structures such as small roads and isolated cars, which results in visually inconsistent maps. In contrast, FSSFFormer (p) yields clearer object outlines and more homogeneous regions across impervious surfaces, buildings, and vegetated areas. The boundaries between adjacent classes are better aligned with the ground truth, and small objects remain identifiable without being oversmoothed. These observations are consistent with the quantitative BF-score improvements reported in Table 3 and support the conclusion that the proposed frequency-domain enhancement enables more precise delineation of fine structures and complex urban patterns.

4.4.2. Results on the ISPRS Potsdam dataset

Table 4 presents the quantitative results on the ISPRS Potsdam dataset. FSSFFormer achieves the highest mIoU of 83.86%, surpassing CDCNet (83.06%) by 0.80 percentage points and BEDSN (82.31%) by 1.55 percentage points. This improvement trend is consistent across other global metrics, with FSSFFormer reaching the best AF (93.37) and OA (91.92%). The performance advantage is more pronounced when examining boundary quality: FSSFFormer obtains a BF-score of 86.95%, which is notably higher than CDCNet (83.48%) and FsaNet (82.17%). Compared with the Vaihingen dataset, the BF-score improvement on Potsdam is even more substantial, indicating that the proposed frequency-domain modeling is particularly effective for high-resolution scenes with sharper edges and more densely distributed small objects.

From a class-wise viewpoint, FSSFFormer provides significant gains in categories where high-frequency structures and fine contours are crucial. The model achieves the best scores on ‘Low vegetation’ (88.63%) and ‘Car’ (97.61%). Compared with CDCNet, the improvements are 0.97 percentage points for ‘Low vegetation’ (87.66% vs. 88.63%) and a remarkable 2.83 percentage points for ‘Car’ (94.78% vs. 97.61%). The Potsdam dataset contains numerous small cars and detailed vegetation patches, where errors often occur at object boundaries or in spectrally ambiguous areas. The superior performance in these categories demonstrates the ability of FSSFFormer to capture multi-frequency cues that are essential for distinguishing subtle spatial transitions.

Although CDCNet slightly outperforms FSSFFormer in ‘Impervious surfaces’ (94.64% vs. 94.42%) and ‘Tree’ (89.53% vs. 88.29%), these differences are minor and occur in classes that are already close to performance saturation. More importantly, CDCNet exhibits noticeable weaknesses in boundary-sensitive classes such as ‘Car’ and fine-scale vegetation. This imbalance leads to lower BF-score and mIoU compared with FSSFFormer. In contrast, our model provides a more uniform improvement across both easy and hard categories, suggesting that its spectral-spatial fusion strategy enhances generalization rather than improving only selected regions.

The boundary-focused evaluation highlights a clear trend on Potsdam. Models like CLCFormer and MSEONet achieve competitive mIoU but show reduced BF-scores (83.02% and 81.64%, respectively), revealing inconsistencies near boundaries and narrow structures. FSSFFormer improves edge alignment and fine-scale contours simultaneously, as reflected by its BF-score of 86.95%. This improvement is larger than that observed on the Vaihingen dataset, indicating that Potsdam’s high spatial resolution and object density amplify the benefit of explicitly modeling frequency subbands. In such scenarios, low-frequency components

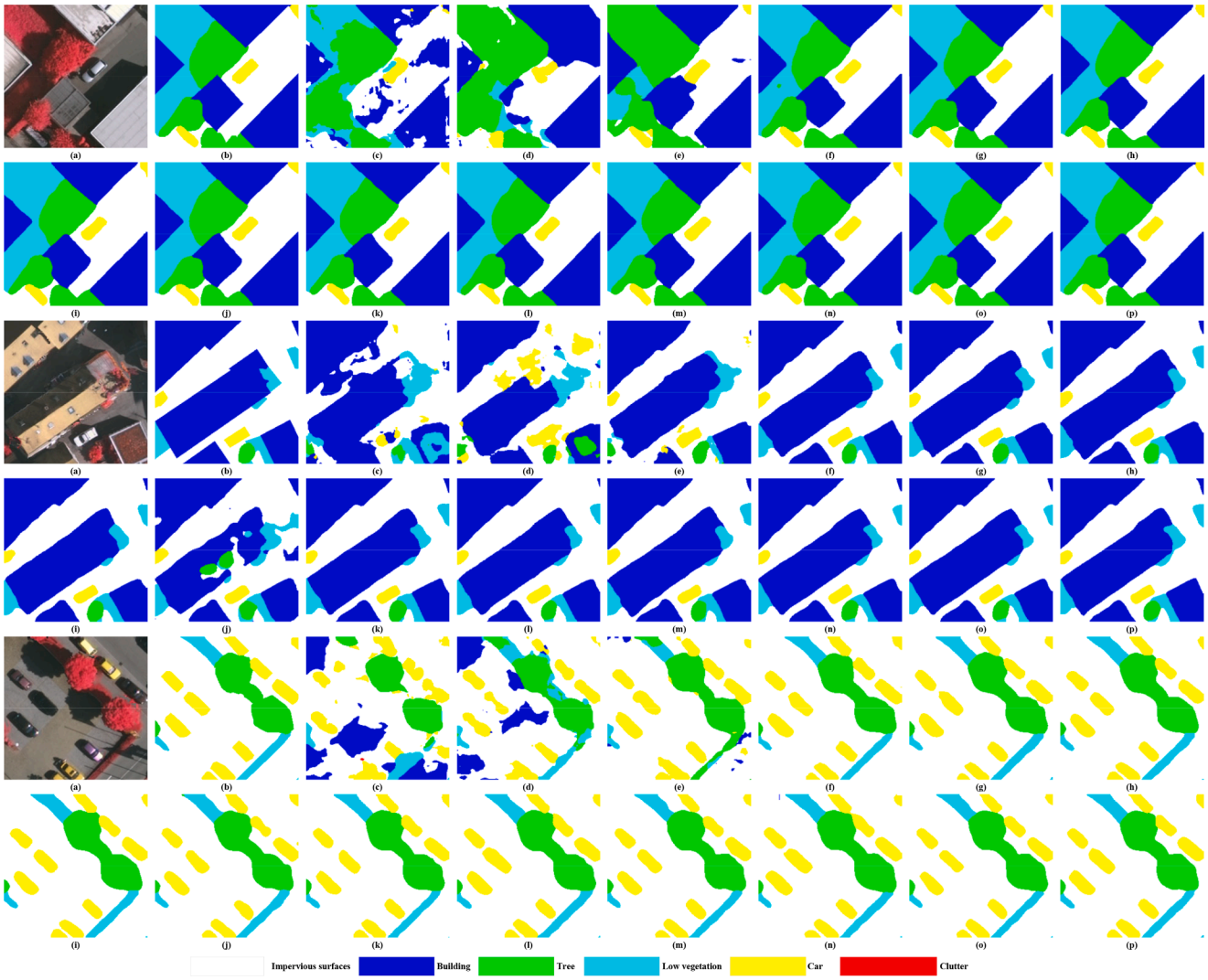


Fig. 7. Visualization comparisons on the ISPRS Vaihingen test set, (a) input, (b) ground truth, (c) U-Net [5], (d) DeepLab V3+ [6], (e) DANet [37], (f) ResUNet-a [16], (g) RAANet [38], (h) MACU-Net [39], (i) SCAttNet [18], (j) A2FPN [19], (k) CLCFormer [20], (l) FsaNet [10], (m) BEDSN [40], (n) CDCNet [32], (o) MSEONet [23], and (p) FSSFFormer (ours).

Table 4
Results on the ISPRS Potsdam Dataset. Class-wise F_1-score, AF, OA, mIoU, and BF-score are listed. Bold indicates the best results.

Methods	Impervious surfaces	Building	Low vegetation	Tree	Car	AF	OA	mIoU	BF-score
U-Net [5]	87.51	89.32	73.98	87.03	48.19	77.21	75.83	70.02	72.11
DeepLabV3+ [6]	84.49	86.13	77.37	77.37	85.47	82.17	80.82	73.59	74.86
DANet [37]	86.96	92.04	80.14	79.95	89.36	85.69	83.99	77.13	78.52
ResUNet-a [16]	85.50	87.15	78.29	78.29	86.49	83.14	81.40	74.11	76.04
RAANet [38]	90.40	95.70	87.20	81.80	77.60	86.54	85.21	77.95	80.73
MACU-Net [39]	89.16	92.76	86.80	83.02	79.25	86.20	85.90	77.99	81.12
SCAttNet [18]	91.87	96.90	85.24	87.05	92.78	90.77	89.06	80.83	83.02
A2FPN [19]	90.67	95.69	85.54	85.44	91.55	89.78	88.25	80.67	82.47
CLCFormer [20]	92.66	96.64	87.05	88.00	93.69	91.61	89.97	81.68	83.02
FsaNet [10]	93.18	98.28	86.46	88.29	94.10	92.06	90.61	81.79	82.17
BEDSN [40]	93.89	97.93	86.33	88.09	94.21	92.09	90.71	82.31	82.91
CDCNet [32]	**94.64**	**97.95**	87.66	**89.53**	94.78	92.91	91.60	83.06	83.48
MSEONet [23]	93.76	96.72	85.39	87.22	91.53	90.52	89.67	81.33	81.64
FSSFFormer (ours)	94.42	97.88	**88.63**	88.29	**97.61**	**93.37**	**91.92**	**83.86**	**86.95**

provide semantic stability, while high-frequency components improve the delineation of object shapes, road margins, and complex vegetation geometries.

Overall, the results confirm that FSSFFormer is well-suited for dense urban scenes, where structural precision and fine-grained boundary alignment play a crucial role. The consistent improvements across global and boundary-sensitive metrics verify that the proposed frequency-aware attention mechanism effectively integrates multi-frequency cues, thereby enhancing both region accuracy and geometric fidelity in the Potsdam dataset.

Fig. 8 provides qualitative comparisons on the ISPRS Potsdam dataset. The Potsdam scenes contain sharper object boundaries, denser

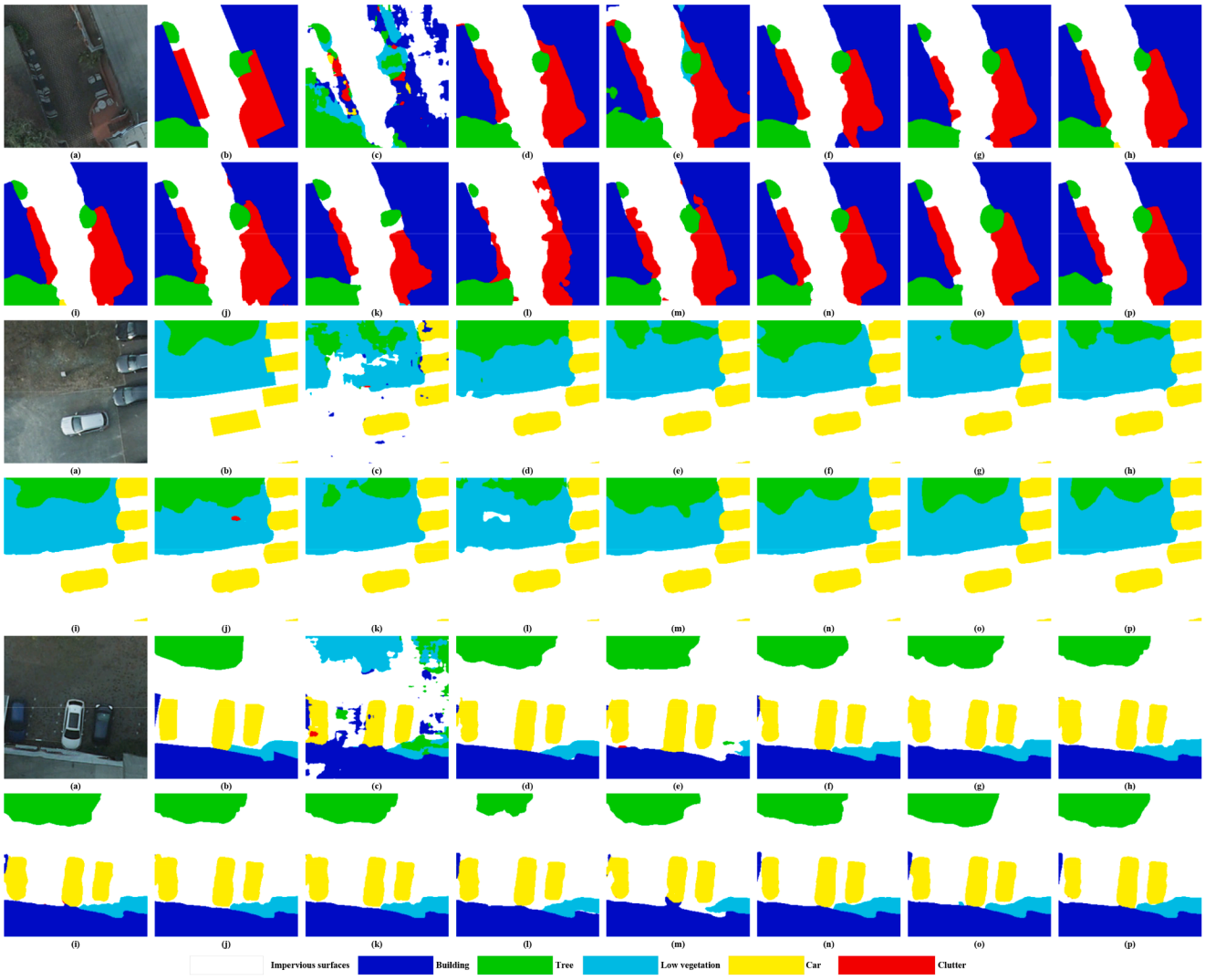


Fig. 8. Visualization comparisons on the ISPRS Potsdam test set, (a) input, (b) ground truth, (c) U-Net [5], (d) DeepLab V3+ [6], (e) DANet [37], (f) ResUNet-a [16], (g) RAANet [38], (h) MACU-Net [39], (i) SCAttNet [18], (j) A2FPN [19], (k) CLCFormer [20], (l) FsaNet [10], (m) BEDSN [40], (n) CDCNet [32], (o) MSEONet [23], and (p) FSSFFormer (ours).

car instances, and more intricate building outlines than Vaihingen, making boundary preservation particularly challenging. Early CNN-based models such as U-Net (c) and DeepLabV3+ (d) exhibit fragmented predictions in car clusters and blurred transitions around building edges. Models incorporating attention mechanisms, including RAANet (g) and SCAttNet (i), improve structural consistency but still fail to recover thin roads and narrow vegetation gaps.

More advanced hybrid architectures such as CDCNet (n) and MSEONet (o) demonstrate stronger region-level accuracy but still produce boundary distortions, especially around small cars, building corners, and detailed vegetation structures. Their predictions often contain broken contours or over-smoothed edges, which lead to misalignment with the ground truth. These issues are reflected in their lower BF-scores in Table 4.

In contrast, FSSFFormer (p) generates the most spatially coherent and geometrically accurate segmentation maps. Car instances remain well separated even in dense parking regions, and building footprints are restored with precise right-angle corners. Vegetation boundaries appear smoother and more consistent, reducing false transitions between 'Tree' and 'Low vegetation'. These improvements match the significantly higher BF-score achieved by our model and highlight the advantages of integrating frequency-domain decomposition with spectral-spatial attention. By selectively enhancing informative frequency bands, FSSFFormer effectively captures both global scene structure and local geometric detail, resulting in superior visual fidelity in complex urban scenes.

4.4.3. Results on the LoveDA dataset

The LoveDA dataset poses a challenging benchmark for semantic segmentation due to its significant intra-class variations, rural-urban domain shift, and diverse scene compositions. As shown in Table 5, FSSFFormer achieves the best overall performance with a mIoU of 67.90%, outperforming CDCNet (66.40%) and BEDSN (66.71%) by 1.50 and 1.19 percentage points, respectively. The gains are consistent in terms of AF and OA, where FSSFFormer attains 78.31 and 74.02%, both of which are the highest among all compared methods. In addition, our model reaches the best BF-score of 72.35%, which is clearly higher than CDCNet (69.02%) and MSEONet (68.11%). This indicates that FSSFFormer not only improves region-based accuracy, but also yields more precise boundaries under complex cross-domain conditions.

From the class-wise perspective, FSSFFormer sets new highs in six out of seven categories. The model delivers F_1-scores of 73.12% on Background, 83.56% on Road, 58.85% on Barren, 71.21% on Forest, and 87.63% on Agriculture, all of which are higher than those of CDCNet.

Table 5
Results on the LoveDA Dataset. Class-wise F_1-score, AF, OA, mIoU, and BF-score are listed. Bold indicates the best results.

Methods	Background	Building	Road	Water	Barren	Forest	Agriculture	AF	OA	mIoU	BF-score
U-Net [5]	50.21	54.74	56.38	77.12	18.09	48.93	66.05	53.07	51.81	47.84	50.26
DeepLabV3+ [6]	52.29	54.99	57.16	77.96	16.11	48.18	67.79	53.50	52.30	47.62	50.03
DANet [37]	54.47	61.02	63.37	79.17	26.63	52.28	70.02	58.14	54.64	50.18	54.91
ResUNet-a [16]	54.15	56.94	59.19	80.73	16.68	49.89	70.20	55.40	53.22	48.46	52.13
RAANet [38]	55.02	62.19	65.58	81.03	29.25	54.11	74.07	60.18	58.95	53.93	58.41
MACU-Net [39]	59.16	64.08	66.73	81.01	32.23	55.81	75.79	62.12	59.65	54.16	60.27
SCAttNet [18]	65.95	71.88	77.04	86.61	50.79	61.19	82.00	70.78	67.31	61.09	64.32
A2FPN [19]	65.17	73.32	75.19	88.01	48.82	59.96	79.71	70.03	66.89	61.14	64.85
CLCFormer [20]	67.17	74.34	77.69	87.71	52.34	64.91	80.96	72.16	70.45	62.55	67.02
FsaNet [10]	68.63	75.95	79.38	89.61	53.47	66.32	82.71	73.72	70.59	63.49	68.05
BEDSN [40]	72.33	**80.18**	84.37	92.14	56.62	69.02	86.47	77.02	73.39	66.71	69.41
CDCNet [32]	71.54	77.16	82.76	92.05	57.25	70.68	86.24	76.81	72.55	66.40	69.02
MSEONet [23]	69.12	76.14	81.83	91.32	54.21	67.45	85.38	75.35	71.26	65.11	68.11
FSSFFormer (ours)	**73.12**	78.58	**83.56**	**93.68**	**58.85**	**71.21**	**87.63**	**78.31**	**74.02**	**67.90**	**72.35**

The improvement is particularly evident for Barren and Road. Compared with CDCNet, FSSFFormer improves Barren from 57.25% to 58.85% and Road from 82.76% to 83.56%. These categories typically exhibit subtle appearance differences and low-contrast boundaries, especially in rural scenes, where misalignment and small boundary errors can significantly degrade mapping quality. The consistent advantage across these difficult classes suggests that the proposed frequency-domain enhancement helps to disentangle texture patterns and refine class transitions.

Compared with recent transformer-based methods such as CLCFormer and CDCNet, FSSFFormer introduces an additional frequency dimension via 2D discrete wavelet transform (DWT), enabling the separation of low-frequency semantic structures and high-frequency edge cues. CLCFormer already offers strong performance with a mIoU of 62.55% and AF of 72.16, yet it lags behind in categories like Barren and Agriculture, and obtains a lower BF-score of 67.02%. CDCNet further improves region-level metrics, but its BF-score remains at 69.02%, indicating that boundaries and thin structures are still not optimally captured. In contrast, FSSFFormer raises both mIoU and BF-score simultaneously, reflecting that the model benefits from frequency-aware attention rather than only relying on spatial transformer blocks.

MSEONet also demonstrates competitive performance on LoveDA, achieving a mIoU of 65.11% and AF of 75.35%. However, it still trails behind FSSFFormer in homogeneous regions such as Forest and Water, and in structurally complex categories such as Road. The lower BF-score of 68.11% for MSEONet indicates that its predictions tend to be less accurate at class boundaries and in areas with high structural complexity. By modeling frequency-decomposed interactions within the transformer framework, FSSFFormer produces segmentation maps that are more robust to domain shift and better aligned with the fine-grained spatial patterns present in both rural and urban scenes.

Overall, the LoveDA results show that the proposed FSSFFormer not only achieves the best global metrics but also delivers markedly improved boundary quality in a cross-domain setting. The consistent advantages over strong CNN-based, transformer-based, and frequency-aware baselines validate that frequency-aware spectral-spatial fusion is an effective strategy for improving generalization under large-scale, heterogeneous remote sensing scenarios.

Fig. 9 provides qualitative comparisons on the LoveDA test set. Due to the rural-urban domain shift, the same semantic class may exhibit very different textures and spatial arrangements. For example, Road regions in rural areas are often narrow and low contrast, while in urban areas they appear wider and more regular. Classical models such as U-Net (c) and DeepLabV3+ (d) frequently produce broken roads, incomplete building footprints, and noisy predictions in Barren regions, especially in rural scenes with sparse textures and complex backgrounds.

Attention-based and transformer-based methods, including SCAttNet (i), CLCFormer (k), CDCNet (n), and MSEONet (o), alleviate some of these issues by introducing global context. Their segmentation maps are more coherent than early CNN baselines, but boundary errors and fine-structure omissions remain visible. In particular, thin roads, small building clusters, and transitions between Forest and Agriculture are often misaligned with the ground truth. These visual artifacts are consistent with their lower BF-scores reported in Table 5, indicating that global context alone is insufficient to fully resolve cross-domain boundary ambiguities.

In contrast, FSSFFormer (p) produces visibly cleaner and more continuous structures across both rural and urban scenes. Road networks are better connected, small buildings are more completely segmented, and Barren and Agriculture areas exhibit fewer spurious fragments. The boundaries between Building, Road, and Background are also more sharply defined. These qualitative improvements align with the superior BF-score and mIoU of FSSFFormer and highlight the benefit of integrating frequency-domain decomposition with spectral-spatial attention. By jointly leveraging low-frequency semantic consistency and high-frequency edge information, FSSFFormer achieves more robust and geometrically accurate segmentation in the presence of strong domain variations.

4.5. Ablation study

4.5.1. Numerical performance

To more comprehensively examine the contributions of the proposed frequency-domain components, we perform an ablation study by selectively removing the SEA and SSCA modules and by replacing both with a standard Multi-Head Self-Attention (MHSA) block. Unlike conventional ablations that only report region-level metrics, we additionally evaluate BF-score to capture boundary fidelity. This four-metric evaluation (AF / OA / mIoU / BF) allows us to assess both semantic consistency and geometric precision across all datasets.

Table 6 summarizes results on Vaihingen, Potsdam, and LoveDA. The full model consistently achieves the best performance across all four metrics, demonstrating that both SEA and SSCA are necessary to fully exploit multi-frequency representations generated by the DWT decomposition. The inclusion of BF-score provides deeper insights: while AF and OA measure overall classification consistency, BF highlights how frequency-domain modeling sharpens boundaries and improves fine-structure recovery.

Removing SSCA leads to the most noticeable performance degradation, particularly in mIoU (-3.35 on Vaihingen; -3.29 on Potsdam; -2.66 on LoveDA) and BF-score (-3.11, -3.32, -2.78 respectively). SSCA models cross-frequency interactions critical for reconstructing coherent structures. Its removal results in misalignment along class boundaries, especially in vegetation clusters (Vaihingen), building footprints (Potsdam), and rural-urban transitions (LoveDA). These observations confirm that long-range spectral-spatial correlation is essential for robust segmentation.

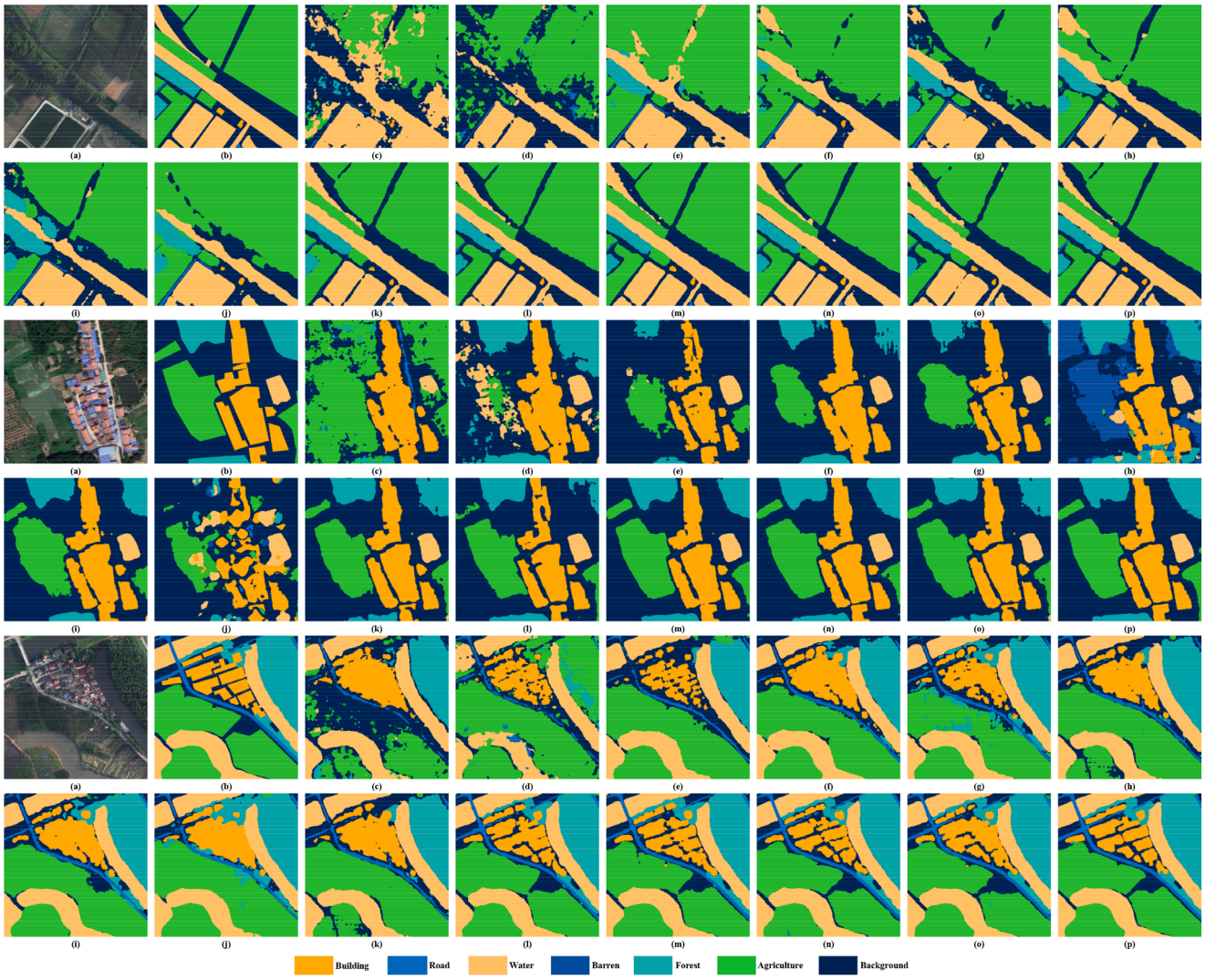


Fig. 9. Visualization comparisons on the LoveDA test set, (a) input, (b) ground truth, (c) U-Net [5], (d) DeepLab V3+ [6], (e) DANet [37], (f) ResUNet-a [16], (g) RAANet [38], (h) MACU-Net [39], (i) SCAttNet [18], (j) A2FPN [19], (k) CLCFormer [20], (l) FsaNet [10], (m) BEDSN [40], (n) CDCNet [32], (o) MSEONet [23], and (p) FSSFFormer (ours).

Removing SEA also reduces performance but to a lesser extent. The BF-score drops more than mIoU across all datasets, reflecting SEA's role in enhancing high-frequency spectral cues that sharpen class boundaries and small objects. This behavior is consistent with the fact that SEA selectively amplifies frequency channels that contribute to fine-grained discrimination, especially relevant in high-resolution datasets.

Replacing both SEA and SSCA with vanilla MHSA leads to the most severe decline in all metrics. Although MHSA captures global context, it lacks inductive bias toward frequency-localized information. As a result, it cannot leverage the DWT subband structure, causing blurred boundaries and inconsistent region predictions. The BF-score degradation is especially pronounced (Vaihingen: -5.79; Potsdam: -5.68; LoveDA: -4.23), highlighting the insufficiency of spatial-only attention mechanisms for frequency-aware remote sensing segmentation.

Overall, the ablation analysis reveals that SEA and SSCA play distinct yet synergistic roles: SEA enhances discriminative frequency responses, while SSCA models inter-band dependencies to reconstruct coherent spatial structures. Their combination yields strong improvements across all four metrics (AF, OA, mIoU, BF), and the consistent gains across datasets further validate the generalizability of our frequency-domain transformer design.

4.5.2. *Efficiency analysis*

To complement the ablation study and quantify the computational footprint of the proposed modules, we measure the number of parameters, floating-point operations (FLOPs), and inference latency for all FSSFFormer variants. All models are evaluated on 256×256 input patches using the same hardware configuration to ensure fairness. The results are summarized in Table 7.

The full model equipped with both SEA and SSCA modules requires 58.50 Mb parameters and 48.27 GFLOPs, with an inference latency of 20.1 ms. Although this configuration incurs the highest computational overhead, it offers substantial performance advantages across all datasets, particularly in boundary accuracy and high-frequency texture modeling.

Removing SSCA reduces parameters and FLOPs by approximately 8%, yielding a moderate speed-up. This reduction is expected because SSCA computes cross-frequency correlation matrices, which contribute a nontrivial portion of the overall cost. However, the corresponding degradation in segmentation quality-especially BF-score-demonstrates that SSCA's modeling capability is worth its computational expense.

Similarly, removing SEA slightly decreases computational cost. SEA is lightweight by design, involving per-frequency spectral reweighting

Table 6
Ablation study on the contributions of SEA and SSCA across three datasets. Results are reported as AF / OA / mIoU / BF (%).

Variant	Submodules			Datasets (AF / OA / mIoU / BF)		
	SEA	SSCA	MHSA	Vaihingen	Potsdam	LoveDA
FSSFFormer (Full)	✓	✓	✗	**92.12 / 91.21 / 83.45 / 84.71**	**93.37 / 91.92 / 83.86 / 86.95**	**78.31 / 74.02 / 67.90 / 72.35**
w/o SSCA	✓	✗	✗	88.77 / 86.83 / 80.10 / 81.60	90.14 / 88.62 / 80.57 / 83.63	75.18 / 72.93 / 65.24 / 69.57
w/o SEA	✗	✓	✗	89.74 / 87.80 / 81.09 / 82.12	91.06 / 89.13 / 81.74 / 84.51	74.66 / 72.54 / 66.05 / 70.11
MHSA (replace both)	✗	✗	✓	87.42 / 85.66 / 78.92 / 78.92	88.17 / 86.75 / 78.65 / 81.27	71.22 / 69.48 / 63.77 / 68.12

Table 7
Efficiency comparison of FSSFFormer variants in terms of parameter count (Mb), FLOPs (G), and average inference time per 256×256 patch (ms).

Variant	Submodules			Params (Mb)	FLOPs (G)	Inference Time (ms ± std)
	SEA	SSCA	Vanilla MHSA			
FSSFFormer (Full)	✓	✓	✗	**58.50**	**48.27**	**20.1 ± 2.2**
w/o SSCA	✓	✗	✗	53.65	45.82	19.0 ± 1.3
w/o SEA	✗	✓	✗	53.98	46.04	20.0 ± 1.5
MHSA (replace both)	✗	✗	✓	50.84	43.76	18.7 ± 1.1

Table 8
Effects of subband decomposition on AF, OA, mIoU, and BF-score across three datasets. LL-only uses low-frequency components only, H-only uses all high-frequency subbands, and Full denotes the proposed FSSFFormer using all subbands.

Methods	Vaihingen				Potsdam				LoveDA			
	AF	OA	mIoU	BF	AF	OA	mIoU	BF	AF	OA	mIoU	BF
LL-only	87.42	89.18	79.12	78.04	89.63	90.52	80.41	79.33	72.54	70.81	62.13	60.55
H-only (LH + HL + HH)	84.75	87.04	75.63	81.92	86.71	88.41	77.08	83.14	70.14	68.90	59.42	66.21
Full (LL + H; ours)	92.12	91.21	83.45	84.71	93.37	91.92	83.86	86.95	78.31	74.02	67.90	72.35

without heavy matrix interactions. Its removal yields only marginal computation savings but produces noticeable performance degradation, confirming its importance in emphasizing subtle yet discriminative frequency responses.

Replacing both frequency-domain modules with standard MHSA yields the lowest computational footprint: 50.84 Mb parameters, 43.76 GFLOPs, and the fastest inference at 18.7 ms. However, as demonstrated in the ablation study, this variant experiences the sharpest performance drop. MHSA lacks structural bias toward frequency decomposition and therefore cannot effectively exploit the multi-band representations produced by the DWT, leading to weaker boundary delineation and inconsistent predictions.

Taken together, the efficiency analysis reveals a clear trade-off: the full FSSFFormer model introduces moderate computational overhead but achieves significantly stronger segmentation performance. Given that remote sensing applications often prioritize accuracy-especially for fine-grained object boundaries and large-scale mapping-the additional cost is justified and remains well within practical deployment limits.

4.6. Effects of subband decomposition

To further investigate the role of frequency decomposition in FSSFFormer, we conduct controlled experiments by selectively enabling different wavelet subbands. Specifically, we evaluate three variants: (1) using only the low-frequency subband (LL-only), (2) using only the aggregated high-frequency subbands (H-only: LH + HL + HH), and (3) using all subbands (Full: LL + LH + HL + HH), which corresponds to the proposed design. These variants are tested on the ISPRS Vaihingen, ISPRS Potsdam, and LoveDA datasets using the same training settings. The results reported in Table 8 provide a comprehensive analysis of how different subbands contribute to segmentation accuracy and boundary fidelity.

Three important trends emerge from the results. First, LL-only yields stable but suboptimal performance across all datasets. Although low-frequency components encode broad semantic structures, their lack of localized contour and texture information limits the model's ability to delineate regions with fine boundary detail. This is reflected in consistently lower BF-scores, especially in LoveDA (60.55%) and Vaihingen (78.04%), where object shapes and vegetation boundaries are highly irregular.

Second, H-only performs notably better in BF-score than LL-only on all datasets, confirming that high-frequency subbands play a critical role in capturing boundary geometry, small objects, and fine-scale spatial details. However, the mIoU and OA values are considerably lower than those of the full model. This suggests that relying solely on high-frequency cues leads to unstable region predictions and inconsistent semantic grouping, since high-frequency information alone is insufficient for modeling large-scale spatial coherence.

Finally, the Full (LL + H) configuration consistently achieves the highest AF, OA, mIoU, and BF-score on all three datasets. The improvements over the LL-only and H-only variants indicate that semantic consistency and structural precision must be jointly modeled to obtain optimal segmentation quality. Across datasets, the Full model improves BF-score by an average of 5.3 points over LL-only and 3.2 points over H-only, highlighting the complementary nature of low- and high-frequency information. The largest BF improvement occurs on Potsdam (LL-only: 79.33%, Full: 86.95%), which contains dense small objects and complex man-made boundaries, further demonstrating the advantages of frequency-aware modeling.

These findings validate the design choice of integrating all DWT subbands in FSSFFormer. By jointly leveraging low-frequency structural cues and high-frequency contour details, the model achieves robust performance across datasets with different spatial resolutions, scene complexities, and domain characteristics.

4.7. Robustness to random initialization

To assess the robustness of FSSFFormer with respect to random initialization and stochastic training dynamics, we perform a multi-seed evaluation on all three datasets. For each dataset, we fix the official train/validation/test split and independently train the model three times with different random seeds. All hyperparameters, data augmentations,

Table 9
Robustness of FSSFFormer across three independent runs with different random seeds. Results are reported as mean ± standard deviation for AF, OA, mIoU, and BF-score on the ISPRS Vaihingen, ISPRS Potsdam, and LoveDA datasets.

Datasets	AF (%)	OA (%)	mIoU (%)	BF-score (%)
ISPRS Vaihingen	92.10 ± 0.12	91.19 ± 0.10	83.42 ± 0.15	84.68 ± 0.18
ISPRS Potsdam	93.35 ± 0.10	91.90 ± 0.09	83.84 ± 0.14	86.92 ± 0.17
LoveDA	78.28 ± 0.20	73.99 ± 0.22	67.87 ± 0.18	72.31 ± 0.21

and optimization settings are kept identical across runs. We then report the mean and standard deviation of AF, OA, mIoU, and BF-score over the three runs, as summarized in Table 9.

Several observations can be drawn from Table 9. First, the standard deviations of mIoU across all three datasets remain below 0.2 percentage points, indicating that the performance of FSSFFormer is highly stable with respect to random initialization. Similar trends are observed for AF, OA, and BF-score, where the variations are consistently small. This suggests that the improvements reported in the comparative experiments are not due to lucky random seeds but reflect the intrinsic capability of the proposed architecture.

Second, the LoveDA dataset exhibits slightly larger variance than the two ISPRS datasets, which is expected given its stronger rural-urban domain shift and higher scene heterogeneity. Even in this challenging setting, the fluctuations remain modest, and the mean performance is still clearly superior to that of competing methods. This demonstrates that the frequency-aware spectral-spatial modeling in FSSFFormer provides not only higher average accuracy but also stable convergence behavior under different random seeds.

Overall, the multi-seed evaluation confirms that FSSFFormer delivers consistent and reliable performance across diverse datasets. The combination of multi-frequency decomposition and transformer-based attention contributes to robust optimization dynamics, making the model suitable for real-world deployment scenarios where reproducibility and stability are critical.

5. Conclusion

In this work, we presented FSSFFormer, a frequency-domain enhanced transformer framework designed to address persistent challenges in fine-grained semantic segmentation of remote sensing imagery. By decomposing inputs into wavelet subbands and introducing two dedicated attention modules, Spectral Enhancement Attention (SEA) and Spectral–Spatial Context Attention (SSCA), the proposed architecture explicitly models the interaction between frequency-sensitive representations and long-range spatial context. This frequency-aware formulation enables the network to better preserve high-frequency boundaries, maintain structural consistency, and improve discrimination in spectrally ambiguous regions, addressing key limitations observed in conventional spatial-domain CNN- and transformer-based approaches.

Extensive experiments on three heterogeneous benchmarks (ISPRS Vaihingen, ISPRS Potsdam, and LoveDA) demonstrate the effectiveness and robustness of the proposed method. Beyond improvements in AF, OA, mIoU, and the newly incorporated boundary-specific BF-score, the expanded experimental analyses-including subband decomposition studies, module-wise ablations, and multi-seed evaluations-provide converging evidence that the observed gains are attributable to the proposed frequency-aware design rather than dataset-specific tuning or random initialization. In particular, the subband and boundary-focused analyses substantiate the role of high-frequency components in improving structural accuracy, while consistent trends across datasets with markedly different acquisition settings and annotation criteria support the architectural generality of the approach.

Efficiency analyses further indicate that FSSFFormer achieves a favorable balance between segmentation accuracy and computational cost. Although the inclusion of SEA and SSCA introduces moderate overhead, these components yield substantial performance benefits, justifying their integration. The inverse DWT reconstruction mechanism is also shown to be essential, enabling a coherent frequency-to-spatial learning loop that preserves fine details without compromising global semantic coherence.

While direct cross-dataset “train-on-one, test-on-another” evaluation is confounded by mismatched label definitions, spatial resolutions, and annotation protocols, the proposed framework itself is not tied to any specific benchmark. In practical remote sensing applications, FSSFFormer can be deployed on new regions or sensors following standard fine-tuning or domain-adaptation practices, which is consistent with common operational workflows. Future work will therefore explore adaptive wavelet learning, improved cross-domain generalization strategies, and extensions to multimodal imagery, such as multispectral, hyperspectral, and SAR data.

Overall, this study establishes a principled and extensible foundation for integrating frequency-domain modeling into transformer architectures for remote sensing segmentation. We expect that the insights presented here will encourage further research on frequency-aware attention mechanisms and their role in improving robustness and structural fidelity in large-scale geospatial understanding tasks.

CRediT authorship contribution statement

Xin Li: Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Formal analysis, Conceptualization; **Feng Xu:** Writing – original draft, Supervision, Project administration, Investigation, Funding acquisition; **Jiaxin Li:** Validation, Software, Formal analysis, Data curation; **Yuanchao Su:** Writing – original draft, Formal analysis, Data curation, Conceptualization; **Linyi Li:** Validation, Supervision, Software, Resources, Data curation; **Xin Lyu:** Project administration, Methodology, Investigation, Funding acquisition; **Zhennan Xu:** Writing – original draft, Visualization, Validation, Software, Formal analysis, Data curation; **André Kaup:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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